

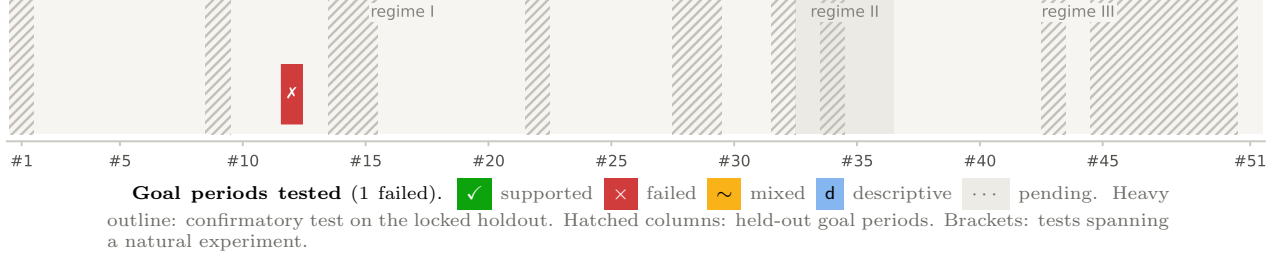
H21: The debate week (#12) is a two-sublattice antiferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

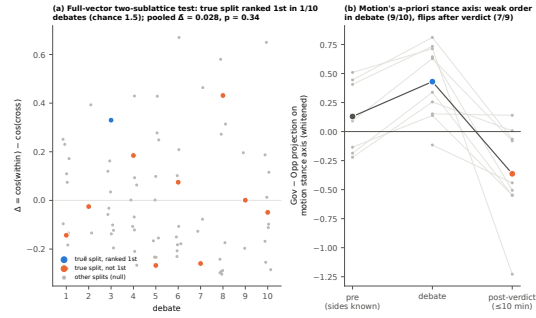
Bottom line. The debate week is a ferromagnet in the motion's topic with family co-variation and only a faint one-dimensional staggered tilt along the stance axis, not a two-sublattice antiferromagnet.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Tested on #12: 10 debates, each with its own judge, motion and re-drafted teams, derived from the record; predictions dated before the run. **The pre-registered two-sublattice test failed:** within- minus cross-team cosine of masked, whitened, agent-centred content vectors $\bar{\Delta} = 0.028$ ($p = 0.34$); the true split was the best split in 1/10 debates (chance 1.5); same-lab pairs out-align team-mates. Order exists only along the motion's *a-priori* support-minus-oppose axis: Government sits on the support side ($\sigma = 0.092$, $p = 0.002$, 9/10), a field about 0.2 of statement noise. The motion topic is a uniform field, off after the verdict (8/8). Post hoc, the stance tilt *reverses* after the verdict (7/9). #34 holdout: not run.



Observables. (a) Per debate, within-minus-cross cosine for every possible team split (gray) and the true split (large dot; blue if best): the true split is unremarkable, so no two-sublattice order in full content vectors. (b) On the motion's a-priori stance axis Government sits slightly on the support side during debates (9/10) and, post hoc, swings to the other side after the verdict (7/9).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly: assigned teams, the strongest public faction structure the village ever had, are invisible as factions in embedding space; topic and model family swamp side. Detecting hidden teams from generic embeddings looks hopeless, and the #34 saboteur test is a long shot. Useful warning: clustering agent messages finds labs and topics, not coalitions. What carries signal is a hand-built axis for the question, which is ordinary stance detection. The post-verdict reversal hints that role-play stances do not persist (post hoc).

PHYSICAL MODEL

Models 11, 10, 01. Spin $\mathbf{s}_i$: debater i 's masked, whitened ($n = 32$), agent-centred chat mean in a debate. Sublattices $A = \text{Government}$ ($\varepsilon_i = +1$), $B = \text{Opposition}$. Fields: motion topic $h_u \hat{g}$ (uniform), assigned sides $h_s \hat{a}$ (staggered), family $\mathbf{h}_i$, verdict h_v toward winner W . Couplings k_{in} (team ferro), k_{out} (cross antiferro). ρ : correlation of sublattice fluctuations.

$$H = -\frac{k_{\text{in}}}{N} \sum_{i \sim j} \mathbf{s}_i \cdot \mathbf{s}_j + \frac{k_{\text{out}}}{N} \sum_{A \times B} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i (h_u \hat{g} + h_s \varepsilon_i \hat{a} + h_v \varepsilon_W \hat{a} + \mathbf{h}_i) \cdot \mathbf{s}_i$$

$$\bar{\Delta} = \langle \cos \rangle_{\text{within}} - \langle \cos \rangle_{\text{cross}}, \quad \rho = -k_{\text{out}} / (1 - k_{\text{in}})$$

H21-debate-antiferromagnet · generated 2026-10-03 19:00 by infra/summaries/build_summaries.py · card hypotheses/H21-debate-antiferromagnet/README.md · ratings: H21 agent (suggested)

RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

G12 $\times$ failed (coupled two-sublattice antiferromagnet);
descriptive: weak one-dimensional staggered...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (**writeup/paper.tex**).

CAVEATS

Pending.

ROUND 2

- **Essence.** LLM agents don't form coalitions; assigned roles leave no lasting imprint.
- **H21-R1** Role imprint lifetime: roles bias stance only while active, then reverse (HH127); measure across all role periods.
- **H21-R2** Agreement is the default stance (sycophancy as ferromagnetic stance coupling); disagreement appears only when assigned.